VisionGuard AI

Detailed Business Requirements Document (BRD)

Enterprise Factory Safety Monitoring & Compliance Platform

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1. Document Information

Item	Value
Project Name	VisionGuard AI
Project Type	AI Powered Factory Safety Monitoring Platform
Prepared For	Factory Management, Safety Department, IT Department
Deployment Model	On-Premise
Platform	Web Based
Target Users	Safety Officers, Supervisors, Plant Managers, Security Teams
Initial Capacity	50 Cameras
Future Capacity	200+ Cameras

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2. Current Business Challenges

Existing Process

Currently factory safety monitoring is performed manually by:

- Supervisors
- Security Teams
- Safety Officers

Monitoring activities include:

- PPE Verification
- Occupancy Monitoring
- Restricted Area Monitoring
- Safety Audits
- Incident Reporting

Problems Identified

Problem 1 – PPE Non Compliance

Workers often enter production areas without:

- Helmet
- Safety Vest
- Safety Shoes
- Gloves

Current detection depends on manual observation.

Issues:

- Human error
- Delayed detection
- No evidence storage
- No audit trail

Problem 2 – Overcrowding

Certain work zones have occupancy limits.

Example:

Zone	Max Capacity
Welding Area	5
Paint Shop	3
Boiler Area	2
Chemical Storage	1

Current Challenges:

- No real-time monitoring
- No automatic alerts
- Safety risk increases

Problem 3 – Delayed Incident Response

Incidents are reported after occurrence.

Examples:

- Worker falls

- Unauthorized entry
- PPE violations

Management receives information too late.

Problem 4 – No Centralized Monitoring

Multiple CCTV systems exist but:

- No centralized dashboard
- No AI analytics
- No alert automation

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3. Proposed Solution

VisionGuard AI will:

- Monitor all factory cameras
- Analyze video streams using AI
- Detect violations automatically
- Generate real-time alerts
- Maintain historical records
- Provide analytics and reporting

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4. Business Goals

Goal 1

Reduce PPE violations by 80%.

Goal 2

Reduce safety incidents by 50%.

Goal 3

Provide real-time monitoring for all production zones.

Goal 4

Improve audit compliance readiness.

Goal 5

Provide evidence-based incident management.

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5. Factory Structure

Factory

A factory contains multiple zones.

Example:

Factory: Greater Noida Plant

Zones

Factory contains:

- Assembly Area
- Welding Area
- Paint Shop
- Packaging Area
- Warehouse
- Dispatch Area
- Loading Dock

Cameras

Each zone can have:

- One camera
- Multiple cameras

Example:

Assembly Area

Camera 1 Camera 2 Camera 3

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6. User Roles

Plant Admin

Responsibilities:

- Manage factories
- Manage users
- Configure zones
- Configure cameras

Safety Officer

Responsibilities:

- Review alerts
- Investigate incidents
- Generate reports

Supervisor

Responsibilities:

- Monitor assigned zones
- Acknowledge alerts
- Take corrective actions

Viewer

Responsibilities:

- View dashboards only

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7. Functional Requirements

Module A - Factory Management

Create Factory

Fields:

Factory Name

Factory Code

Address

Plant Head

Status

Edit Factory

Admin can modify:

- Factory Details
- Status

Module B - Zone Management

Create Zone

Fields:

Zone Name

Zone Code

Occupancy Limit

Supervisor

Description

Example

Zone:

Welding Area

Occupancy Limit:

5

Supervisor:

Pankaj Gupta

Module C - Camera Management

Add Camera

Fields:

Camera Name

Camera Code

RTSP URL

Zone

Status

Camera Health Monitoring

System shall monitor:

- Camera Online
- Camera Offline
- Stream Delay
- FPS

Camera Failure Rule

IF Camera Offline > 60 Seconds

THEN Generate Alert

Severity = High

Module D - PPE Detection

Helmet Detection

System shall detect:

Helmet Present

Helmet Missing

Detection Flow

Camera Frame

↓

Person Detection

↓

Helmet Detection

↓

Violation Evaluation

Business Rule

IF

Person Detected

AND

Helmet Missing

THEN

Create Violation

Evidence Storage

Store:

- Snapshot
- Timestamp
- Camera
- Zone

Module E - Safety Vest Detection

System shall verify safety vest compliance.

Rule:

IF

Vest Missing

THEN

Create Violation

Module F - Occupancy Monitoring

System shall continuously count persons.

Occupancy Rule

IF

Current Count > Occupancy Limit

THEN

Generate Overcrowding Alert

Example

Zone:

Welding Area

Limit:

5

Current Count:

8

Result:

Overcrowding Alert

Module G - Alert Management

Alert Categories

Critical

- Fire
- Explosion
- Fall Detection

High

- Helmet Missing
- Overcrowding
- Restricted Area Entry

Medium

- Vest Missing

Low

- Camera FPS Drop

Module H - Incident Management

Every violation becomes an incident.

Incident Fields:

Incident Number

Incident Type

Zone

Camera

Timestamp

Severity

Status

Snapshot

Incident Status

Open

Acknowledged

In Progress

Closed

Module I - Dashboard

Overview Dashboard

Display:

Total Cameras

Online Cameras

Offline Cameras

Active Alerts

Open Incidents

PPE Compliance %

Zone Dashboard

Display:

Current Occupancy

Maximum Occupancy

Current Violations

Zone Health

Camera Dashboard

Display:

Camera Name

Status

FPS

Stream Health

Module J - Analytics

PPE Analytics

Display:

Helmet Compliance %

Vest Compliance %

Violations by Zone

Violations by Month

Occupancy Analytics

Display:

Peak Occupancy

Average Occupancy

Overcrowding Frequency

Safety Score

Calculate:

Compliance %

Incident Rate

Alert Closure Time

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8. Workflow Scenarios

Scenario 1

Helmet Missing

Worker enters zone.

AI detects person.

AI detects no helmet.

System creates violation.

System stores image.

System sends alert.

Supervisor receives notification.

Scenario 2

Overcrowding

Maximum Occupancy = 5

Current Count = 7

System detects violation.

Alert generated.

Supervisor notified.

Violation logged.

Scenario 3

Camera Offline

Camera disconnected.

System fails heartbeat check.

After 60 seconds:

Alert generated.

IT team notified.

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9. Non Functional Requirements

Performance

Support:

50 Cameras

Future:

200 Cameras

Alert Response

Maximum:

5 Seconds

Availability

99.5%

Security

JWT Authentication

Role Based Access

Audit Logs

HTTPS

Scalability

Horizontal scaling supported.

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10. Integration Requirements

Future integrations:

- SAP
- HRMS
- Attendance System
- Access Control System
- Fire Alarm System
- SMS Gateway
- Email Server

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11. Hardware Requirements

Development

CPU:

8+ Core

RAM:

32 GB

GPU:

RTX 2000 Ada

Production (50 Cameras)

CPU:

24 Core

RAM:

64 GB

GPU:

RTX 4090

Storage:

2 TB SSD

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12. KPIs

Helmet Compliance %

Vest Compliance %

Average Occupancy

Incident Count

Alert Response Time

Alert Closure Time

Safety Score

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13. Success Criteria

- Helmet Detection Accuracy ≥ 90%
- Occupancy Accuracy ≥ 95%
- Alert Latency ≤ 5 Seconds
- Camera Uptime Monitoring
- Real-Time Dashboard Available
- Historical Analytics Available
- Support 50 Cameras Simultaneously

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14. Future Enhancements

Phase 2

- Face Recognition

- Visitor Tracking
- Attendance

Phase 3

- Fire Detection
- Smoke Detection
- Fall Detection

Phase 4

- Forklift Monitoring
- Unsafe Behavior Detection
- Predictive Safety Analytics
- AI Risk Scoring
- Digital Safety Audits